Lesson 39: Optimisation Problems — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Differentiate $A = 20x - x^2$.

2. Solve $20 - 2x = 0$.

3. What does setting $f' = 0$ find?

4. To confirm a maximum, f'' should be?

5. To confirm a minimum, f'' should be?

6. Differentiate $A = 16x - x^2$.

Section 2 — Core practice

7. $A = 12x - x^2$. Find x for maximum A, then the maximum.

8. $A = 16x - x^2$. Find x for max A, then the maximum.

9. $C = x^2 - 10x + 30$. Find x for minimum C.

10. $A = 24x - x^2$. Find x for maximum area.

11. $C = x^2 - 6x + 11$. Find x for minimum C.

12. $A = x(20 - x)$. Expand and find x for max.

Section 3 — Challenge & exam-style

13. A farmer has 40 m of fence for three sides of a rectangle, area $A = x(40 - 2x)$. Find x for max area and the area.

14. A rectangle has perimeter 24 cm. Show the area is $A = x(12 - x)$ and find the maximum area.

15. An open box from a square base has volume $V = x^2(10 - x)$. Find x for maximum V ($x > 0$).

16. Cost $C = x^2 - 14x + 60$. Find the minimum cost and the x at which it occurs.